



Outline

- Classes of materials
- Chemistry review
- Review of atomic structure
- Review of the periodic table
- Density, atomic # & wt, mole, Avogadro's #
- Bonding forces and energy



Classes of materials

	POLYMERS	CERAMICS	METALS
DUCTILITY	Varies	Poor	Good
CONDUCTIVITY (ELECTRICAL & THERMAL)	Low	Low	 Poor
HARDNESS/STRENGTH	Low – medium	Very high	Medium – high
CORROSION RESISTANCE	Fair – good		Fair – poor
STIFFNESS	Low	Good	Fair
FRACTURE TOUGHNESS	Low – medium	Low	
MACHINABILITY	Good	Poor / Low	Good



Why study bonding?

- Because the **properties of materials** (*strength, hardness, conductivity, etc..*) **are determined by electronegativity** **which atoms are connected.**
- Also by how the atoms are arranged in space →

What determines the nature of the chemical bond between atoms?

crystal structure

- Electronic structure (distribution of electrons in atomic orbitals)
- Number of electrons and (tendency for an atom to attract an electron)

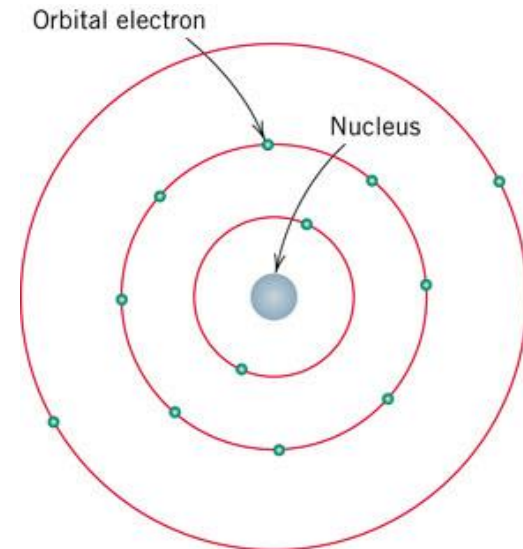


Atomic Structure

Bohr model of the atom: (1913)

– Bohr atomic model

- electrons revolve around nucleus in discrete orbitals

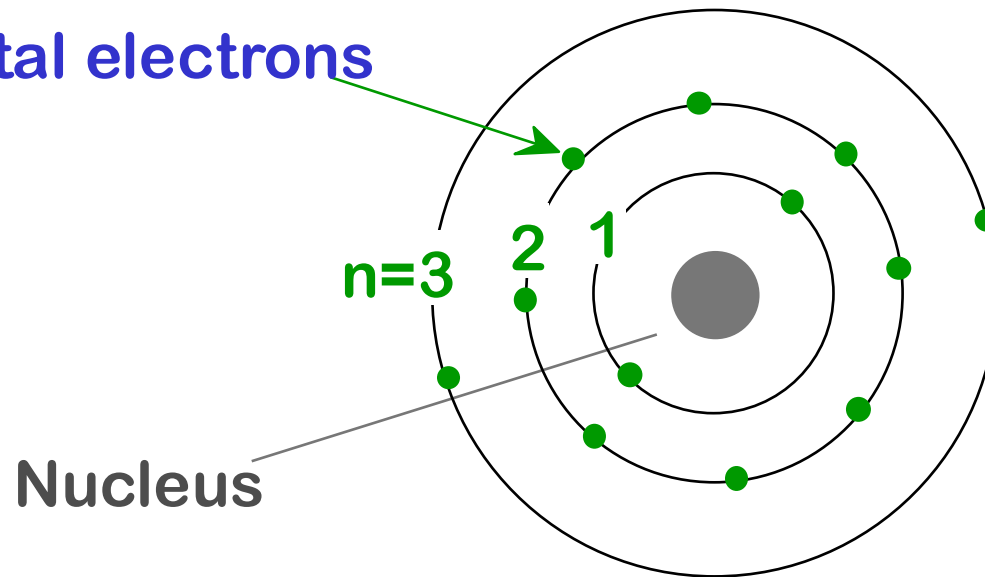


Based on earlier work of Rutherford and his own from spectral emission studies



Bohr Atom

orbital electrons



$Z = \# \text{ protons}$
 $= 1 \text{ for hydrogen}$
 $\text{to } 94 \text{ for plutonium}$

$N = \# \text{ neutrons}$

Atomic mass $A \approx Z + N$

Isotopes: same Z , different $N \rightarrow$ same element, different atomic mass (e.g. ^{12}C and ^{13}C).

Reported atomic mass = abundance-weighted average of the isotopes, e.g. C:
 $0.989(12.000) + 0.011(13.003) \approx 12.011 \text{ g/mol}$



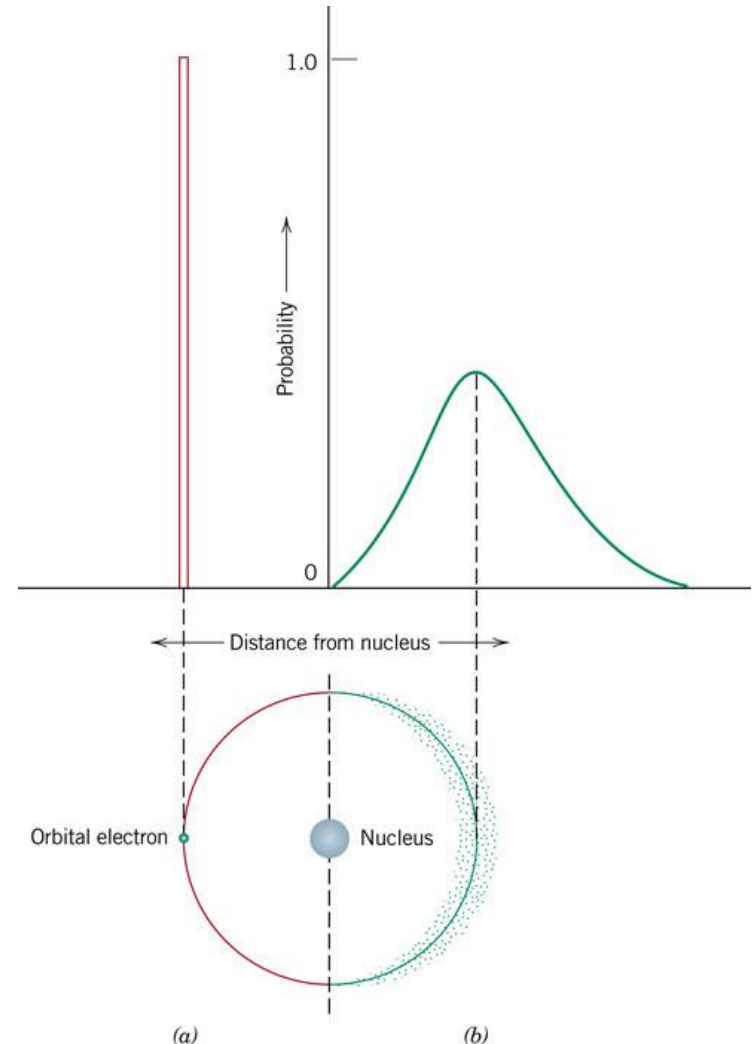
Atomic Structure

Bohr vs. Quantum-Mechanical Model: (1927)

❖ Wave-mechanical atomic model

- Position of electron is imprecisely known; only a probability distribution.
- Electron exhibits both particle and wave characteristics

*Schroedinger, Heisenburg, Planck and others developed this model (**wave mechanics**), which allows a more precise description of the atom.*





1	New Designation
IA	Original Designation

(Mass Numbers in Parentheses are from the most stable of common isotopes.)

Solid
Liquid
Gas

Rare Earth Elements

Lanthanide Series

Actinide Series



The Periodic Table

Electronegativity - each kind of atom has a certain attraction for the electrons involved in a chemical bond. This "**electron-attracting**" power of each atom can be listed numerically on an electronegativity scale.

Electronegativity Values of Selected Elements

Metallic Elements			Nonmetallic Elements			
Li (1.0)	Be (1.5)		C (2.5)	N (3.0)	O (3.5)	F (4.0)
Na (1.0)	Mg (1.2)	Al (1.5)		P (2.1)	S (2.5)	Cl (3.0)
K (0.9)	Ca (1.0)	Sc (1.3)			Se (2.4)	Br (2.8)

*Electronegativity was originally worked out by Linus Pauling in 1939
– see “The Nature of the Chemical Bond”*



- Adapted
from Fig. 2.6,
Callister 8e.

**Electronegative elements:
Readily acquire electrons
to become $-$ ions.**



Electronegativity

- Ranges from **0.7** to **4.0**,
- Large values: tendency to acquire electrons.

IA																																0
1 H 2.1	IIA												IIIA	IVA	VA	VIA	VIIA														2 He -	
3 Li 1.0	4 Be 1.5												5 B 2.0	6 C 2.5	7 N 3.0	8 O 3.5	9 F 4.0													10 Ne -		
11 Na 0.9	12 Mg 1.2												13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	17 Cl 3.0													18 Ar -		
		IIIB	IVB	VB	VIB	VIIIB	VIII			IB	IIB																					
19 K 0.8	20 Ca 1.0	21 Sc 1.3	22 Ti 1.5	23 V 1.6	24 Cr 1.6	25 Mn 1.5	26 Fe 1.8	27 Co 1.8	28 Ni 1.8	29 Cu 1.9	30 Zn 1.6	31 Ga 1.6	32 Ge 1.8	33 As 2.0	34 Se 2.4	35 Br 2.8														36 Kr -		
37 Rb 0.8	38 Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5														54 Xe -		
55 Cs 0.7	56 Ba 0.9	57-71 La-Lu 1.1-1.2	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	85 At 2.2														86 Rn -		
87 Fr 0.7	88 Ra 0.9	89-102 Ac-No 1.1-1.7																														


Smaller electronegativity


Larger electronegativity

Adapted from Fig. 2.7, *Callister 8e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940.



ELECTRONEGATIVITY

- **High electronegativity** → strong tendency to accept an electron (i.e., Group VIIA: F, Cl)
- **Low electronegativity** (called “electropositive”) → strong tendency to give up an electron, i.e., Group IA: Li, Na, K)

The **difference in electronegativity between two atoms determines the resulting *electron distribution* and the *type of bond***



Density, Atomic # & Wt, Mole & Avogadro's

- Density

- g/cm³ (most solids range ~ 1 - 23 g/cm³)

- Atomic number – number of protons (Z)
 6.022×10^{23}

- Atomic weight (A)

- g/mole

$A \cong \text{number protons (Z)} + \text{neutrons (N)}$

$$\cong Z + N$$

- Mole = number of particles

- $N_A = \dots \dots \dots \text{part/mole}$



Review Problems

- How many atoms in 6 grams of carbon?
- Calculate the volume of 1 mole of Au.

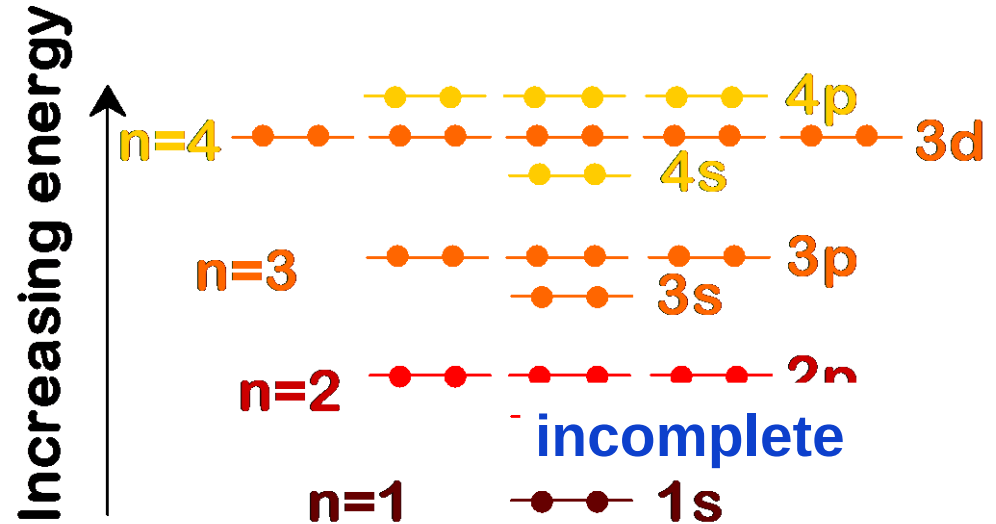
*Useful tip: many problems can be worked by suitable manipulation of density (g/cm^3), atomic mass (g/mole), and Avagadro's number (atoms/mole) (use **dimensional analysis!**)*



Electron Energy States

Electrons...

- have discrete **energy states**
- tend to occupy lowest available energy state.

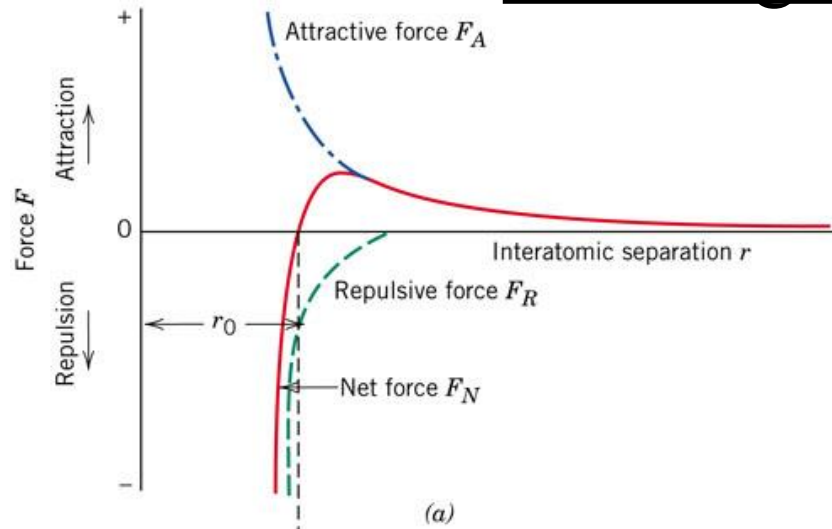


Stable electron configurations...

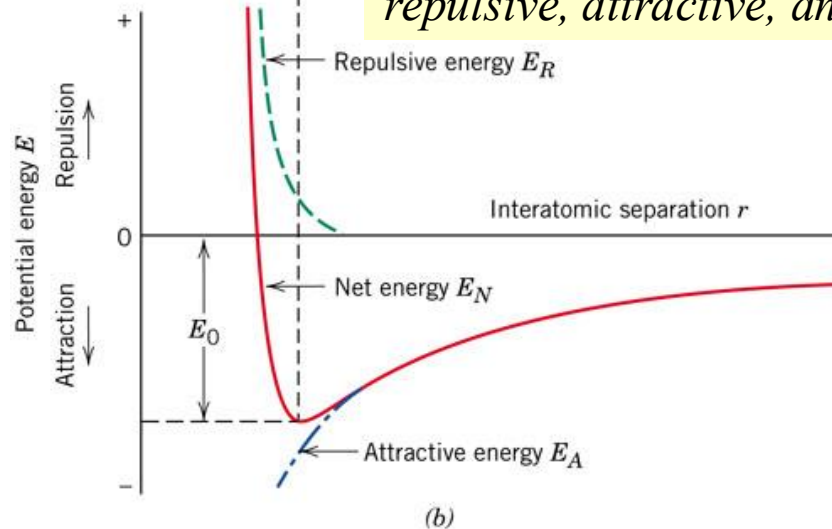
- have complete s and p subshells
 - tend to be
- Most elements: Electric **inert / unreactive**



Bonding Forces and Energies

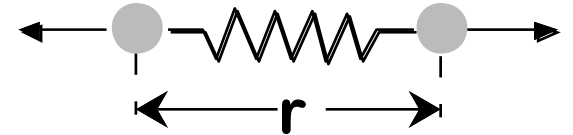


repulsive, attractive, and net forces



repulsive, attractive, and net energies

Bond length, r



Net force is given by the **sum** of an **attractive** force and a **repulsive** force

Potential is given by the integral of the net force curve with respect to distance:

$$E = \int F \cdot dr$$

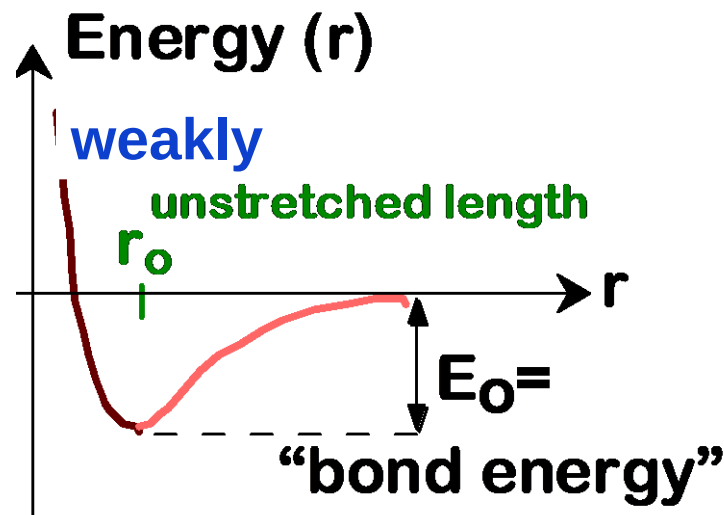
Note: *equilibrium separation occurs where the net force = 0*



Bonding Forces and Energies

Bonding energy: *Minimum of the potential vs. distance curve.*

- Indicates how much energy must be supplied to completely disassociate the two atoms
 - Depth of the potential well indicates bonding strength
 - Deep well → bonded
 - Shallow well → bonded
- strongly**





Bond Energy

The higher the ~~bond energy~~

strong bond energy

-

moderate bond energy

-

weak bond energy

- State as function of bonding energy

- Solid (.....)

- Liquid (.....)

- **Higher melting temperature (T_m)**

- **Higher elastic modulus (E), lower thermal expansion (α)**



Next time:

Types of Atomic Bonds